

Root Locus Theory

3.1 What Root Locus Shows

For a closed-loop system with loop gain $K \cdot G(s)$, the characteristic equation is:

$$1 + K \cdot G(s) = 0$$

The roots of this equation — the closed-loop poles — change as K varies from 0 to ∞ . The root locus is the complete picture of all possible closed-loop pole locations for every value of $K \geq 0$.

Reading a root locus directly gives:

stability (LHP = stable, RHP = unstable), **damping ratio** (angle from negative real axis), **natural frequency** (distance from origin), and **settling time** (real part of pole).

Root Locus Construction Rules

The following rules govern the shape of every root locus:

1. Number of branches equals number of open-loop poles (n). Each branch represents one closed-loop pole trajectory.
2. Branches start ($K=0$) at open-loop poles and end ($K \rightarrow \infty$) at open-loop zeros or along asymptotes to infinity.
3. Number of asymptotes = $n - m$, where m is number of finite zeros.
4. Asymptote angles:

$$\phi_a = \frac{(2k + 1) \times 180^\circ}{n - m} \quad \text{for } k = 0, 1, \dots, n - m - 1$$

5. Asymptote centroid:

$$\sigma_a = \frac{\sum \text{Poles} - \sum \text{Zeros}}{n - m}$$

6. Real axis: locus exists to the left of an odd number of real poles and zeros.

7. Breakaway and break-in points occur where:

$$\frac{dK}{ds} = 0$$

Converting Performance Specs to Pole Locations

Time-domain performance specifications translate directly to desired pole locations in the complex plane:

1. Damping Ratio (ζ) from Percent Overshoot (**OS**%)

$$\zeta = \frac{-\ln(OS/100)}{\sqrt{\pi^2 + \ln^2(OS/100)}}$$

For $OS < 10\%$: $\zeta \geq 0.59$

2. Real Part of Pole (σ) from Settling Time (t_s)

$$\sigma = \frac{4}{t_s}$$

For $t_s < 10s$: $\sigma \geq 0.4$

These two requirements define a triangular region in the left-half plane — bounded by the damping ratio lines and the settling time vertical line — where the closed-loop dominant poles must be located for the specifications to be satisfied.

Aircraft Pitch Plant

The transfer function from Project 04 is used without modification. With the positive-gain sign convention matching the Simulink closed-loop implementation:

$$G(s) = \frac{-1.282 + 1.282s}{s^3 + 1.935s^2 + 0.987s + 0.179}$$

Table 1 Aircraft Pitch Plant Parameters

Parameter	Value	Physical Meaning
K (positive convention)	+1.282	Positive pitch for positive elevator command
Zero location	$s = +1$ (RHP)	Non-minimum phase — undershoot
Pole p_1	-1.2679	Short Period mode — fast, heavily damped
Pole $p_{2,3}$	$-0.3336 \pm 0.1730j$	Phugoid mode — slow, lightly damped, dominant